

Machine Learning

Course Syllabus

Fall 2026 • 3 Credit Hours

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1 Course Description

This course provides a comprehensive introduction to machine learning, spanning the mathematical foundations (linear algebra, optimization, probability estimation), classical machine learning (regression, classification, kNN, decision trees, clustering, dimensionality reduction), modern deep learning (artificial neural networks, autoencoders, convolutional neural networks, recurrent neural networks), and advanced research topics (explainable AI, active learning, and physics-informed neural networks). The course emphasizes both theoretical understanding and hands-on implementation using Python, NumPy, PyTorch, and modern ML libraries. Students complete a semester-long project applying course concepts to a real dataset.

2 Prerequisites

- **Linear Algebra:** matrix operations, eigenvalues, SVD
- **Calculus:** partial derivatives, chain rule, gradients
- **Probability & Statistics:** distributions, Bayes' theorem, MLE, MAP
- **Programming:** Python (NumPy, Matplotlib); basic PyTorch helpful

3 Course Learning Outcomes

Upon successful completion of this course, students will be able to:

- CLO1. Formulate** machine learning problems using linear algebra, optimization, and probability.
- CLO2. Implement** core ML algorithms (regression, classification, kNN, decision trees, k-means, PCA, neural networks) from scratch.
- CLO3. Estimate** probabilistic models using MLE, MAP, and Bayesian data fusion.
- CLO4. Build** deep learning models (ANN, autoencoder, CNN, RNN) using PyTorch/TensorFlow.
- CLO5. Explain** model predictions using XAI techniques (SHAP, LIME, Grad-CAM, CAM).
- CLO6. Design** active learning pipelines and physics-informed neural networks.
- CLO7. Communicate** technical results through written reports and oral presentations.

4 Textbooks and Resources

- Chatterjee I. Machine Learning and Its Applications: A Quick Guide for Beginners. Sharjah, United Arab Emirates: Bentham Science Publishers; 2021 Dec 22.
- Burkov A. The hundred-page machine learning book. Quebec City, QC, Canada: Andriy Burkov; 2019 Jan 1.

5 Course Schedule

Week	#	Topic	Key Concepts
Part 1 – Mathematical Foundations			
Module 1 – Linear Algebra (Slides 01)			
1	1	Systems & Vectors	System of linear equations, matrix form $Ax = b$, inverse, inner product, vector norms, orthogonality
1	2	Matrices as Linear Maps	Matrix–vector products, linear transformation, rotation, stretch/compress, projection, composition of maps
2	3	Eigenvalues & Eigenvectors	$Av = \lambda v$, eigenvalue analysis, projection, mirror, rotation matrices (2D/3D)
2	4	Linear Systems & SVD	Well-/under-/over-determined systems, least-norm solution, least-squares, orthogonal projection, SVD
Module 2 – Optimization (Slides 02)			
3	5	Optimization Basics	Objective function, decision variables, constraints, convex sets and functions
3	6	Gradient Descent	Gradient, closed-form vs. iterative solutions, learning rate, stopping criteria, convergence
4	7	Convex Problems in Practice	Linear programming, quadratic programming, CVXPY solver
4	8	Applications	Shortest distance, empty bucket, supply chain, de-noising, total variation
Module 3 – Dataset & Probability (Slides Dataset, 23)			
5	9	Datasets	What is a dataset, features, labels, records, data types
5	10	Data Preprocessing	Missing values, duplicates, outliers, encoding categorical data, feature scaling, feature engineering, data reduction
6	11	Probability Density Estimation	Gaussian, MLE for μ and σ^2 , log-likelihood
6	12	Data Fusion & MAP	Two-sensor fusion, recursive Bayesian update, Maximum a Posteriori, Kalman filter, sensor fusion in the brain
Part 2 – Classical Machine Learning			
Module 4 – Regression (Slides 03)			

Week	#	Topic	Key Concepts
7	13	Linear Regression	Linear model, least-squares solution $\theta^* = (\Phi^T \Phi)^{-1} \Phi^T y$, gradient descent solution, CVXPY solution
7	14	L_1 vs. L_2 Regression	Norm-based regression, sensitivity to outliers, robustness of L_1
8	15	Nonlinear Regression	Polynomial regression, RBF basis, linear basis function models, explicit feature vectors
8	16	Regularization	Overfitting, ridge (L2), LASSO (L1), shrinkage, feature selection, bias-variance trade-off
Module 5 – Classification (Slides 04)			
9	17	Perceptron	Linear classification boundary, hyperplane, perceptron update rule, loss function, Python implementation
9	18	Support Vector Machine	Margin maximization, distance to hyperplane, slack variables, soft-margin SVM, kernels, CVXPY implementation
10	19	Logistic Regression	Sigmoid function, maximum likelihood, cross-entropy, gradient descent, CVXPY, scikit-learn
10	20	Multiclass Classification	One-vs-one, one-vs-all, softmax, one-hot encoding, nonlinear classification with kernels
11	—	Midterm Exam	Covers Parts 1–2
Part 3 – Clustering, kNN, Trees, Dim. Reduction			
Module 6 – kNN & Decision Trees (Slides 05)			
12	21	k-Nearest Neighbors	KNN regression, KNN classification, choice of k , distance metric, decision boundaries
12	22	Decision Trees	Feature tests, entropy/disorder, information gain, quality of split, ID3 tree construction, random forest
Module 7 – Clustering (Slides 06)			
13	23	K-Means	Unsupervised learning, within-cluster similarity, k-means algorithm, chicken-and-egg dilemma, convergence
13	24	K-Means in Practice	Initialization sensitivity, elbow method for choosing k , limitations (hard assignments, outliers, non-convex clusters)
Module 8 – Dimensionality Reduction (Slides 07)			
14	25	PCA	Variance maximization, covariance matrix, eigenvectors, principal components, projection
14	26	SVD, POD & FDA/LDA	SVD, low-rank approximation, Proper Orthogonal Decomposition, Eigenfaces, Fisher Discriminant Analysis

Week	#	Topic	Key Concepts
Part 4 – Deep Learning			
Module 9 – Artificial Neural Networks (Slides 08)			
15	27	From Perceptron to MLP	Binary linear classifier, XOR problem, multi-layer perceptron, hidden layers as learned kernels, universal function approximator
15	28	Training Neural Networks	Activation functions (Sigmoid, Tanh, ReLU), backpropagation, chain rule, dynamic programming, TensorFlow implementation (MNIST)
Module 10 – Autoencoders (Slides 09)			
16	29	Autoencoder Basics	Encoder-decoder, latent space, dimensionality reduction, feature extraction, denoising
16	30	Autoencoders in Practice	MNIST autoencoder, latent-space visualization, generative autoencoder, latent-space interpolation, pendulum dynamics, flow around cylinder
Module 11 – Convolutional Neural Networks (Slides 10)			
17	31	Convolution	1D and 2D convolution, kernels, filters, feature maps, padding, stride, multi-channel convolution
17	32	CNN Architecture	Nonlinear activation, pooling (max, average, global), locality, weight sharing, convolution layer block
18	33	CNN in Practice	LeNet, AlexNet, VGG, ResNet, transfer learning, MNIST with TensorFlow, steel surface defect classification
Module 12 – Explainable AI (Slides 11)			
18	34	XAI Fundamentals	Black-box problem, transparency, trust, fairness, feature importance
19	35	XAI Methods	SHAP (game-theoretic), LIME (local surrogate), Saliency Maps (gradient-based)
19	36	Grad-CAM & CAM	Gradient-weighted Class Activation Maps, Global Average Pooling, CAM heatmap, steel surface defects, wind turbine energy prediction
Module 13 – Recurrent Neural Networks (Slides 12)			
20	37	Sequence Modeling	Time-series data, finite vs. infinite response, autoregressive models, state-space models, folded vs. unfolded RNN
20	38	RNN Training	Backpropagation through time (BPTT), vanishing/exploding gradients, word embedding, LSTM, GRU
21	39	RNN in Practice	Vibration signal prediction, MNIST half-image completion, recursive generation

Week	#	Topic	Key Concepts
Part 5 – Advanced and Emerging Topics			
Module 14 – Active Learning (Slides 13)			
22	40	Active Learning Basics	Segmentation, weakly-supervised learning, WSL + active learning, labeling cost reduction
22	41	Sampling Strategies	Uncertainty sampling (least confidence, margin), Query-by-Committee, expected model change, core-set
23	42	Bayesian Optimization	Surrogate models (GPR, BNN), utility functions (PI, UCB, EI), exploration vs. exploitation, hyperparameter tuning
Module 15 – Physics-Informed Neural Networks (Slides 15)			
23	43	PINN Fundamentals	Data-driven vs. physics-informed, taxonomy of informed deep learning, universal approximation, ODEs and PDEs
24	44	PINN Architecture	Physics loss, initial/boundary condition loss, data loss, collocation points, mesh-free approach, automatic differentiation
24	45	PINN Applications	Euler beam, thin plate (Föppl-von Kármán), Navier-Stokes (flow around cylinder), heat transfer, inverse problems
Module 16 – Parameter Estimation (Slides 23)			
25	46	MLE	Probability density estimation, Gaussian MLE, likelihood, log-likelihood
25	47	MAP & Kalman Filter	Bayesian inference, prior, posterior, recursive Bayesian estimation, Kalman filter (prediction/correction)
26	48	Probabilistic ML	Data fusion, sensor fusion, probabilistic regression/classification/clustering/dimension reduction
Review and Assessment			
26	49	Project Presentations	Student project demos and peer feedback
26	50	Final Review	Comprehensive review and Q&A
27	—	Final Exam	Comprehensive (all modules)

6 Assessment and Grading

Component	Description	Weight
Homework Assignments	6 assignments (theory + coding)	25%
Midterm Exam	Covers Parts 1–2 (Week 11)	20%
Final Exam	Comprehensive (Week 27)	25%
Course Project	Personal project with report + presentation	30%
Total		100%

7 Course Policies

Late Submission

Late homework loses 10% per day, up to 3 days. After 3 days, no credit unless prior arrangements are made.

Academic Integrity

All submitted work must be your own. Collaboration on homework is encouraged for discussion, but each student must write and submit their own solutions. Plagiarism results in a zero.

Use of AI Tools

Generative AI tools (e.g., ChatGPT, Copilot) may be used for learning and debugging, but all submitted code and writing must be the student's own. Any use of AI must be disclosed in a brief note at the end of each assignment.

Attendance

Attendance is expected at all lectures. More than 3 unexcused absences may result in a grade penalty.

Penalty

Anyone who arrives late to class must treat all their classmates to ice cream.

8 Software and Tools

- **Python 3.10+** with NumPy, SciPy, Matplotlib, Pandas, scikit-learn
- **Deep learning:** PyTorch (primary), TensorFlow/Keras
- **XAI:** SHAP, LIME, Captum, tf-keras-vis
- **PINN:** DeepXDE, PyTorch, or JAX

End of Syllabus